

HARROW, ON, Canada

WMO: 712980

Lat: 42.033N

Lon: 82.896W

Elev: 192

StdP: 99.04

Time Zone: -5.00 (NAE)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.3	-13.4	-20.6	0.6	-15.8	-17.9	0.8	-12.8	10.8	-5.7	10.0	-5.2	3.5	250	0.483

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	30.7	22.9	29.4	22.3	28.0	21.6	24.6	28.5	23.6	27.4	22.8	26.6	4.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	18.6	26.9	22.5	17.6	26.1	21.6	16.6	25.3	75.4	28.4	71.7	27.4	68.5	26.6	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
9.3	8.2	7.2	DB	-19.5	33.0	2.9	1.6	-21.6	34.2	-23.3	35.1	-25.0	36.0	-27.1	37.2	
			WB	-19.9	26.0	2.8	1.3	-22.0	26.9	-23.6	27.7	-25.2	28.4	-27.2	29.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-3.7	-2.7	2.0	8.4	14.8	20.3	22.5	21.5	18.0	11.6	5.2	-0.8
	DBStd	10.36	5.95	5.34	5.34	4.73	4.59	3.65	3.10	2.85	3.96	4.69	4.67	5.07
	HDD10.0	1656	425	357	254	84	8	0	0	0	1	36	156	335
	HDD18.3	3492	683	590	506	298	129	21	4	6	54	214	394	593
	CDD10.0	1593	0	1	7	37	158	308	388	356	240	87	12	1
	CDD18.3	388	0	0	0	1	21	79	134	103	43	6	0	0
	CDH23.3	2632	0	0	0	8	138	538	1030	662	236	20	0	0
	CDH26.7	608	0	0	0	0	26	145	268	122	46	2	0	0
Wind	WSAvg	3.3	4.2	3.9	3.6	3.7	3.1	2.8	2.5	2.3	2.5	3.2	3.7	4.0
Precipitation	PrecAvg	895	63	58	66	83	88	80	87	81	89	65	68	69
	PrecMax	1413	105	135	142	140	170	134	166	167	185	146	177	138
	PrecMin	710	21	17	16	31	32	30	26	11	26	10	16	31
	PrecStd	143	24	28	29	29	42	33	36	38	48	31	32	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.3	13.6	19.5	24.6	29.3	31.9	32.6	31.4	30.4	25.8	18.3	13.2
		MCWB	10.3	11.1	15.2	17.1	20.8	22.4	23.8	23.7	23.0	20.3	13.9	11.4
	2%	DB	9.1	9.3	15.0	21.5	26.9	30.1	31.0	29.4	27.9	22.9	15.6	10.4
		MCWB	8.1	7.5	11.7	14.9	19.6	21.8	23.4	22.9	21.7	18.5	12.6	9.2
	5%	DB	6.3	6.3	12.6	18.6	24.7	28.4	29.6	27.9	25.9	20.8	13.9	7.9
		MCWB	5.0	4.5	9.9	13.3	18.6	21.0	22.5	22.1	20.4	16.9	11.2	6.5
	10%	DB	3.9	3.9	9.9	16.1	22.3	26.5	28.0	26.7	24.2	18.9	12.1	5.7
		MCWB	2.7	2.2	7.5	11.5	17.2	19.9	21.9	21.5	19.7	15.3	9.8	4.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.5	11.4	16.1	18.0	22.5	24.1	26.5	25.6	24.4	21.1	15.2	11.6
		MCDB	11.1	12.9	19.4	23.0	27.5	29.3	30.7	28.9	29.0	24.4	17.0	12.9
	2%	WB	8.2	7.6	12.4	16.0	21.0	23.1	24.9	24.2	22.8	19.4	13.1	9.3
		MCDB	9.0	9.2	14.3	20.0	24.9	28.1	28.7	27.6	26.1	21.9	15.2	10.3
	5%	WB	5.1	4.5	10.2	14.1	19.5	22.2	23.9	23.3	21.7	17.4	11.5	6.6
		MCDB	5.9	6.0	12.2	17.7	23.1	26.7	27.6	26.6	24.8	20.1	13.4	7.7
	10%	WB	2.8	2.3	7.6	12.2	18.1	21.3	23.0	22.5	20.7	15.7	9.9	4.5
		MCDB	3.8	3.7	9.7	15.6	21.7	25.2	26.7	25.6	23.4	18.4	11.9	5.6
Mean Daily Temperature Range	5% DB	MDBR	6.4	6.7	7.7	9.5	9.5	9.1	9.3	8.9	9.1	8.2	6.8	5.5
		MCDBR	7.7	8.9	10.9	12.6	11.2	10.5	9.9	9.5	10.0	9.8	8.6	7.1
		MCWBR	7.3	7.8	8.4	8.2	6.0	4.7	4.6	4.8	5.2	5.8	6.9	6.7
	5% WB	MCDBR	7.6	8.8	10.3	11.6	10.0	9.3	8.7	7.9	8.5	8.7	8.1	7.2
		MCWBR	7.6	8.1	8.4	8.4	6.3	4.8	4.7	4.6	5.1	5.8	7.1	7.1
Clear-Sky Solar Irradiance	taub	0.305	0.334	0.355	0.398	0.434	0.447	0.438	0.430	0.387	0.361	0.321	0.315	
	taud	2.401	2.299	2.327	2.255	2.201	2.214	2.263	2.274	2.366	2.419	2.493	2.425	
	Ebn at Noon	839	863	888	870	845	832	835	831	844	825	817	793	
	Edh at Noon	79	103	113	131	142	141	133	127	108	91	73	72	
All-Sky Solar Radiation	RadAvg	1.45	2.31	3.47	4.62	5.42	6.09	6.09	5.31	4.29	2.79	1.74	1.21	
	RadStd	0.15	0.28	0.33	0.39	0.46	0.39	0.32	0.21	0.37	0.23	0.23	0.11	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+91	N/A

Nomenclature: See separate page